

Wednesday, May 14, 2025 3:26 PM

|   |                                                                    |
|---|--------------------------------------------------------------------|
| ⑦ | Solvent Clean<br>5-10 minutes in acetone, IPA, methanol, and water |
| ⑧ | Ash <i>O<sub>2</sub> plasma cleaner</i><br>5 minutes               |
| ⑨ | Dehydrate <i>Hot plate</i><br>180C, 5 min                          |
| ⑩ | HMDS<br>105C, 2 min / 60°C, 2 min                                  |
| ⑪ | Spin coat <i>AZ 5214</i><br>5000 rpm, 1 min                        |
|   | Soft bake<br>100C 90s                                              |
|   | Dose [80 104.2] mJ/cm <sup>2</sup><br>Defocus: -2                  |
|   | Post exposure bake<br>100C, 40s                                    |
|   | Flood exposure <i>↔ UV Light Curing</i><br>45s                     |
|   | Develop in MF 319<br>~40s                                          |

<https://dvh.physics.illinois.edu/pdf/AZ5214E.pdf>

80 to 104 mJ/cm<sup>2</sup> in steps of 2

basically the same results with 102 and 104. Sometimes those could start to touch sub-micron resolution, but not reliably.

Issues I faced:

The MLA just won't expose sometimes caused by some bug.

Always revisit:

1. Dose rate
2. Resist and development (time and type)
3. Adhesion promotor (HMDS)

<https://heidelberg-instruments.com/wp-content/uploads/2021/10/fact-sheet-MLA150.pdf>

<https://snfguide.stanford.edu/guide/equipment/heidelberg-mla-150-heidelberg>

### Yuanchen's recipe

AZ 5214 negative resist approach (for metal liftoff)

1. Clean/Asp strip 150W 5mins
2. 180C dehydration bake 5mins
3. HMDs recipe 3@3000rpm
4. 5214 recipe 5@5000rpm
5. 90C 30s softbake
6. M.A, invert if pattern is positive, 40mJ, defoc 3
7. Post bake 110C/90s
8. Flood exposure 45s
9. MF319, 45s (might be longer, check pattern

AZ 5214 negative resist approach (for metal liftoff)

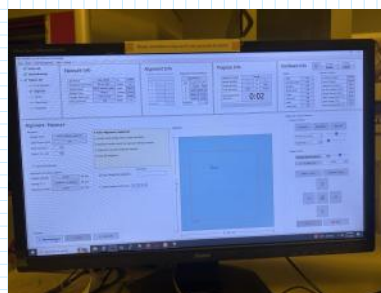
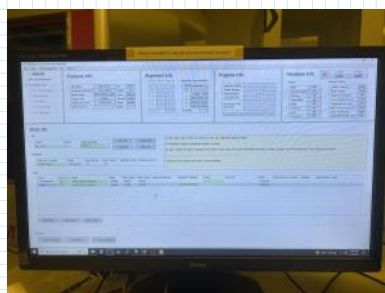
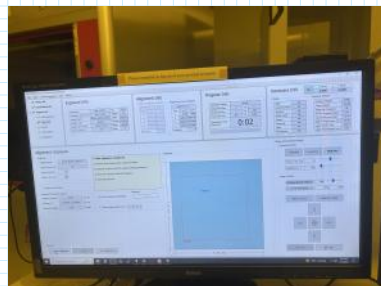
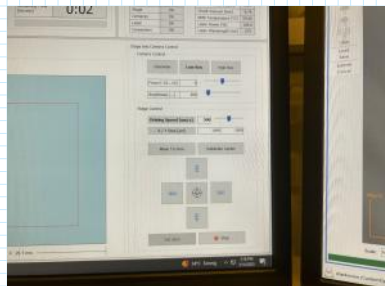
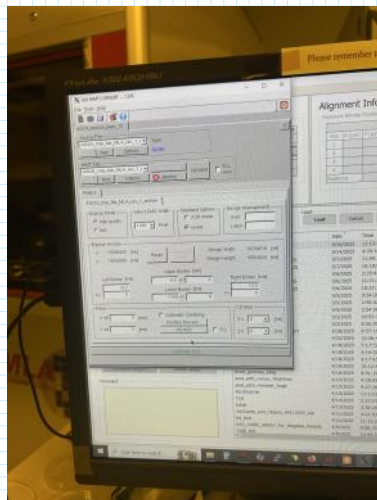
1. Clean/AS strip 150W 5mins
2. 180C dehydration bake 5mins
3. HMDS recipe 3@5000rpm
4. 90C 60s bake
5. 5214 recipe 5@5000rpm → Use cold AZ 5214 in fridge
6. 90C 30s softbake
7. MLA, invert if pattern is positive, 40mJ, defoc value -2 for UA
8. Post bake 110C/90s
9. Flood exposure 45s
10. MF319, 45s (might be longer, check pattern)
11. Wash immediately in DI Water (do not over develop)

Roland shared this document which has the processing guidelines for AZ 5214.

[https://www.microchemicals.com/dokumente/datenblaetter/tds/merck/en/tds\\_az5214e\\_photoresist.pdf](https://www.microchemicals.com/dokumente/datenblaetter/tds/merck/en/tds_az5214e_photoresist.pdf)

In this I see Post-exposure baking (named as Reversal bake) to be 120 degrees C for 2 min.

The recipe I did is 100C for 40s and your recipe (I haven't tried this yet) says 110C for 90s. The document says the post exposure bake is the most critical step of all. Should I do 120C for 2mins this time?



IMG\_8127

Tested Recipe for MLA 150 @ OSC

Positive resist A7 5214E

- Solvent clean [Acetone, IPA, Methylanol, DI Water]
- 180C dehydration 5 mins in hotplate
- Spin 5214 @ 5000 rpm for 3 min [Keep the AZ5214 resist in fridge for storage and leave it outside for a few hours or a day before coating it]
- Bake 110°C for 90 sec
- MLA exposure 350 mJ/cm<sup>2</sup> (no invert) Defocus value : -2
  - 350 mJ/cm<sup>2</sup> dose worked best for positive tone, defocus remains the same
- Developer: MF 319 for 45 seconds and wash with DI water multiple times immediately.
  - Ideal develop is between 45 +/- 5 seconds



MLA150\_  
manual 2...

← Refer the manual for software controls, setup and exposure settings

## Tested Recipe for MLA 150 @ OSC

AZ 5214E Negative resist Image reversal approach (for metal liftoff)

1. Solvent clean [Acetone, IPA, Methanol, DI Water]
2. 180C dehydration 5 mins in hotplate
3. Spin 5214 @ 5000 rpm for 3 min [Keep the AZ5214 resist in fridge for storage and leave it outside for a few hours or a day before coating it]
4. Bake 90°C for 30 sec
5. MLA exposure 40 mJ/cm<sup>2</sup> (invert if pattern is positive),
  - a. Defocus value : -2
  - b. 40 mJ/cm<sup>2</sup> dose worked best for negative tone, defocus remains the same
6. Post exposure Bake 110°C for 90 sec
7. UV Flood exposure 45 sec
8. Developer: MF 319 for 45 seconds and wash with DI water multiple times immediately.
  - a. Ideal develop is between 45 +/- 5 seconds.